



MSC. IN
DATA SCIENCE
Máster Universitario en Ciencia de Datos



Practical application 2: probabilistic supervised classification

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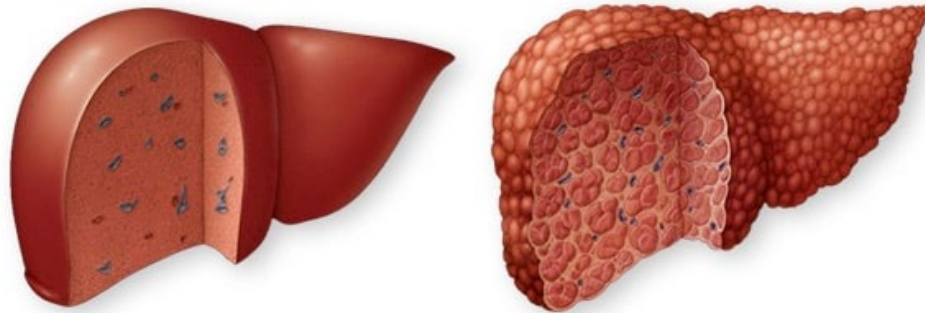
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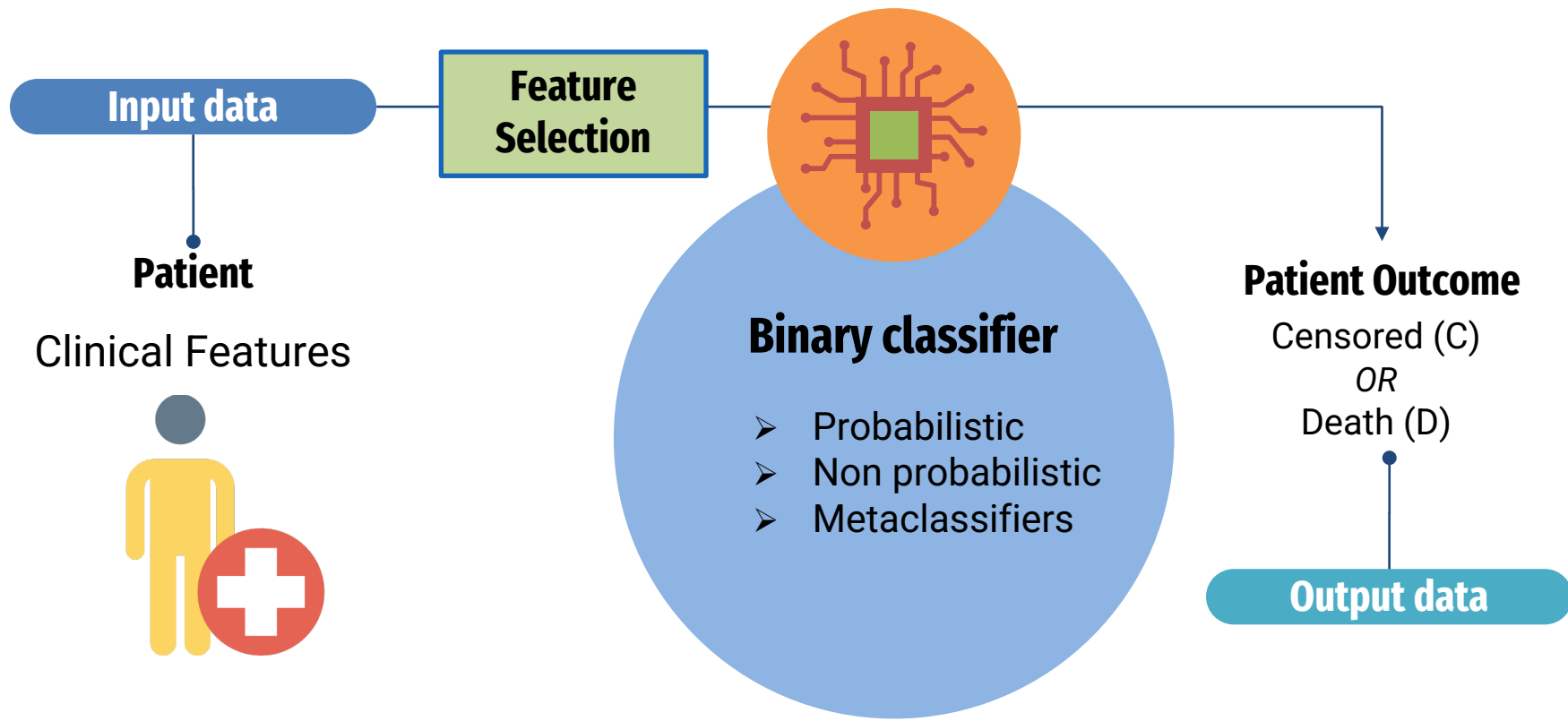
Introduction

Liver cirrhosis is a late stage of scarring (fibrosis) of the liver caused by many forms of liver diseases and conditions, such as hepatitis and chronic alcoholism.



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Introduction



Problem Description

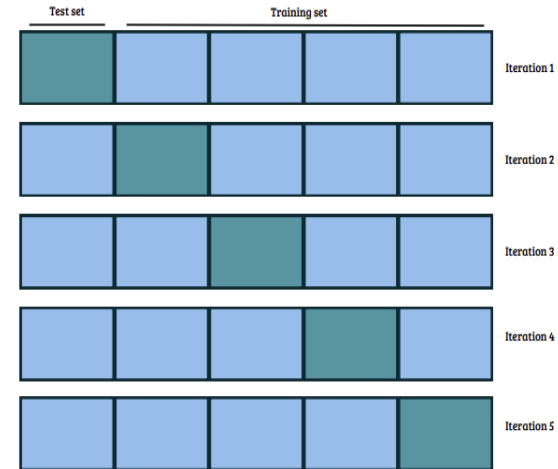
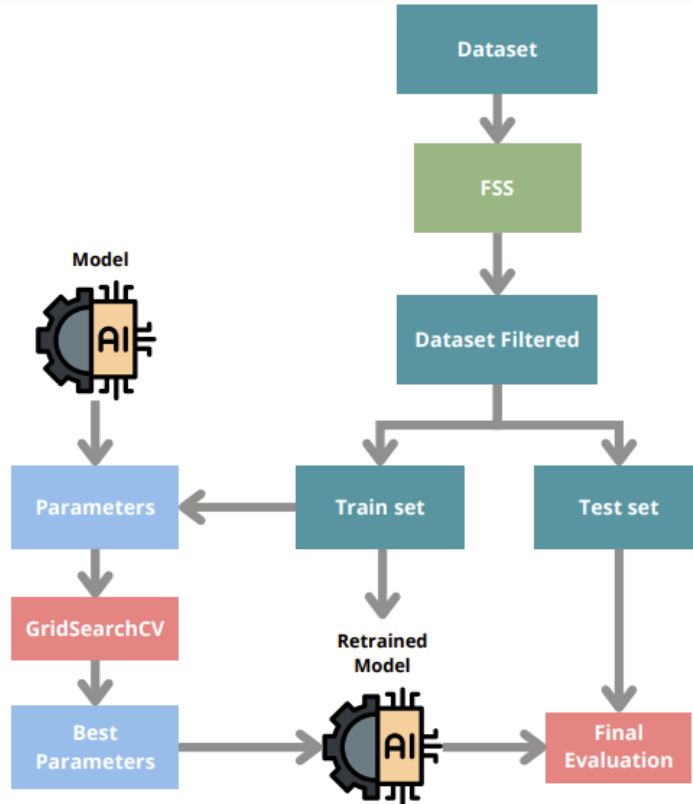
- The dataset contains 293 instances, each representing an individual patient and each one, includes 25 clinical features.
- Patient Outcome (target)
 - 168 (Censored = class 0)
 - 125 (Death == class 1)

Table 1: Data-set Analysis

Feature	Mean	Std	Min	Max	Type
Age	50.6271	10.5698	26.2958	78.4931	continuous
Albumin	3.5169	0.4229	1.96	4.64	continuous
Alk_Phos	2011.6709	2195.9565	289.0	13862.4	continuous
Ascites_N	0.918089	0.274699	0.0	1.0	binary
Ascites_Y	0.081911	0.274699	0.0	1.0	binary
Bilirubin	3.264164	4.648182	0.3	28.0	continuous
Cholesterol	365.2108	212.7557	120.0	1775.0	continuous
Copper	95.9395	84.2078	4.0	588.0	continuous
Drug_D-penicillamine	0.5051	0.5008	0.0	1.0	binary
Drug_Placebo	0.4948	0.5008	0.0	1.0	binary
Edema_N	0.8395	0.3676	0.0	1.0	binary
Edema_S	0.0921	0.2897	0.0	1.0	binary
Edema_Y	0.0682	0.2526	0.0	1.0	binary
Hepatomegaly_N	0.4948	0.5008	0.0	1.0	binary
Hepatomegaly_Y	0.5051	0.5008	0.0	1.0	binary
N_Days	2038.6655	1137.3298	41.0	4556.0	continuous
Platelets	259.5417	95.5279	62.0	563.0	continuous
Prothrombin	10.7488	1.0219	9.0	17.1	continuous
Sex_F	0.8873	0.3166	0.0	1.0	binary
Sex_M	0.1126	0.3166	0.0	1.0	binary
SGOT	122.0661	57.7574	26.35	457.25	continuous
Spiders_N	0.70989	0.4545	0.0	1.0	binary
Spiders_Y	0.2901	0.4545	0.0	1.0	binary
Stage	3.0170	0.8854	1.0	4.0	nominal
Tryglicerides	124.1343	61.5558	44.0	598.00	continuous

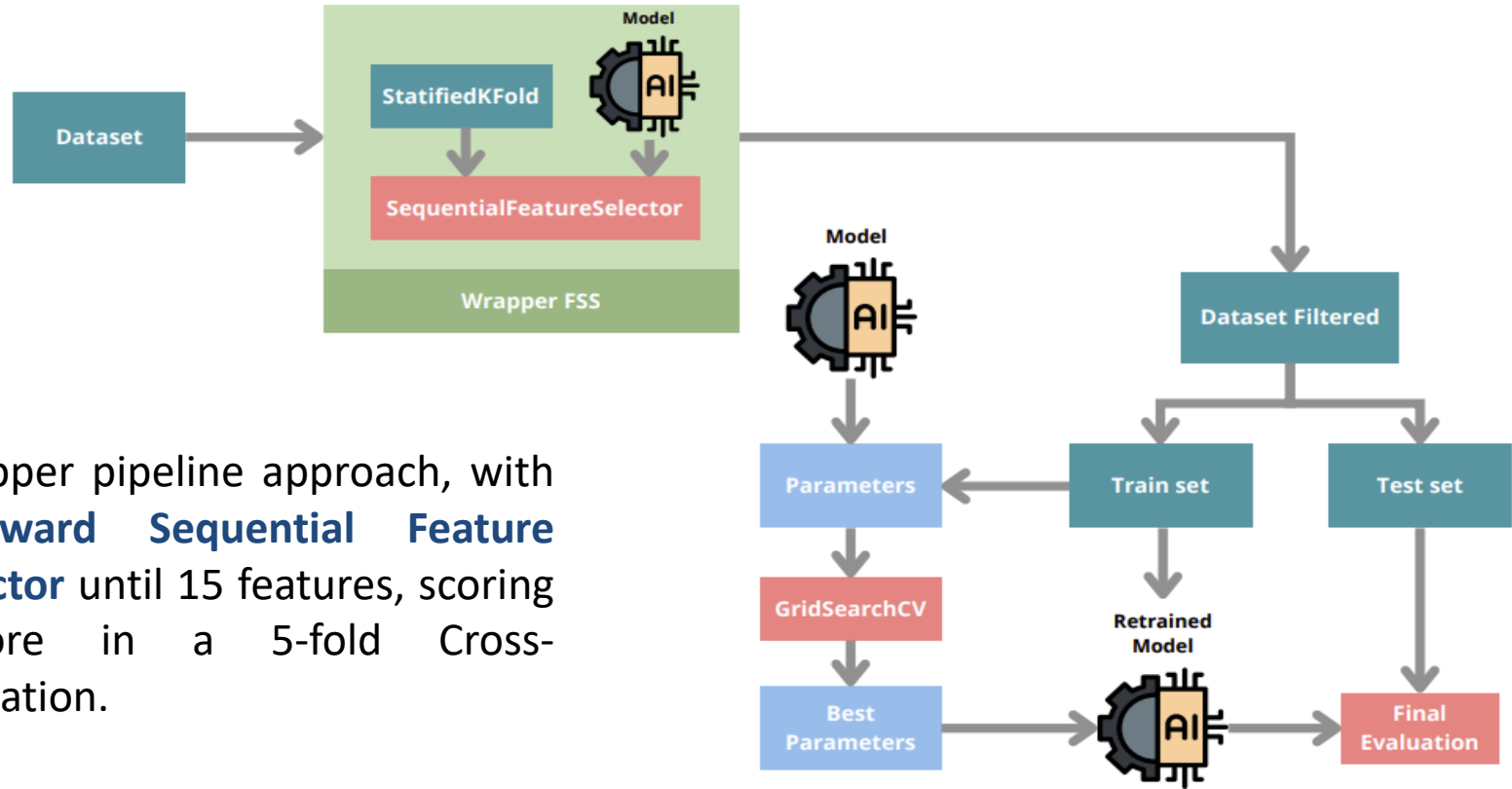
Methodology

- Automated evaluation pipeline for FSS univariate (**Mutual Information**) and multivariate FSS (**Relief**) approach, only 15 features are selected. After get the Dataset filtered, I realize a 5-Cross Validation.



Methodology

- Wrapper pipeline approach, with **Backward Sequential Feature Selector** until 15 features, scoring F-score in a 5-fold Cross-Validation.



Model Selection

- **Bagging:** Decision Tree (unstable classifier).
- **Stacking:** Support Vector Machine and Decision Tree (variability), while Logistic Regression as the final classifier.
- **Voting:** Support Vector Machine, Logistic Regression, and Random Forest (variability).

<i>n estimators</i>	10, 50, 100
<i>max samples</i>	0.5, 1.0
<i>max features</i>	0.5, 1.0
<i>base estimator max depth</i>	3, 5, 10

<i>svm C</i>	0.1, 1
<i>dt max depth</i>	3, 5, 7
<i>final estimator C</i>	0.1, 1

<i>lr C</i>	0.1, 1
<i>svc C</i>	0.1, 1
<i>rf n estimators</i>	10, 50, 100
<i>voting</i>	soft, hard

Table 3: Selected features by each method

Feature	Univariate	Multivariate	Wrapper								
			LDA	QDA	LR	GNB	RF	AB	Bagging	Stacking	Voting
Age	✓		✓	✓	✓		✓	✓	✓	✓	✓
Albumin	✓	✓		✓	✓	✓		✓		✓	
Alk_Phos	✓	✓			✓	✓		✓			✓
Ascites_N	✓		✓				✓		✓	✓	✓
Ascites_Y				✓	✓	✓	✓	✓	✓		✓
Bilirubin	✓	✓	✓	✓		✓	✓		✓	✓	✓
Cholesterol				✓		✓		✓		✓	✓
Copper	✓	✓		✓	✓	✓	✓	✓	✓	✓	
Drug_D-penicillamine	✓			✓			✓		✓		
Drug_Placebo			✓				✓				✓
Edema_N	✓	✓			✓		✓		✓	✓	
Edema_S	✓		✓	✓	✓			✓	✓	✓	✓
Edema_Y	✓		✓				✓	✓	✓	✓	✓
Hepatomegaly_N	✓	✓	✓		✓	✓	✓		✓		✓
Hepatomegaly_Y	✓	✓	✓	✓	✓	✓			✓		
N_Days	✓	✓	✓	✓	✓	✓		✓			✓
Platelets								✓		✓	
Prothrombin	✓	✓	✓	✓		✓	✓	✓	✓	✓	
Sex_F		✓	✓	✓	✓	✓	✓	✓			✓
Sex_M		✓	✓	✓	✓	✓		✓		✓	
SGOT		✓	✓		✓	✓	✓	✓	✓	✓	✓
Spiders_N	✓	✓		✓	✓		✓		✓	✓	✓
Spiders_Y		✓	✓			✓	✓		✓	✓	
Stage	✓	✓		✓		✓					✓
Tryglicerides			✓		✓			✓			

* DT: Decision Tree, MLP: Multi Layer Perceptron; SVM: Supper Vector Machine;
 LDA: Linear Discriminant Analysis; QDA: Quadratic Discriminant Analysis; LR: Logistic Regression;
 GNB: Gaussian Naive Bayes; RF: Random Forest; AB: Ada Boost;

Experiments & Results

Experiments & Results

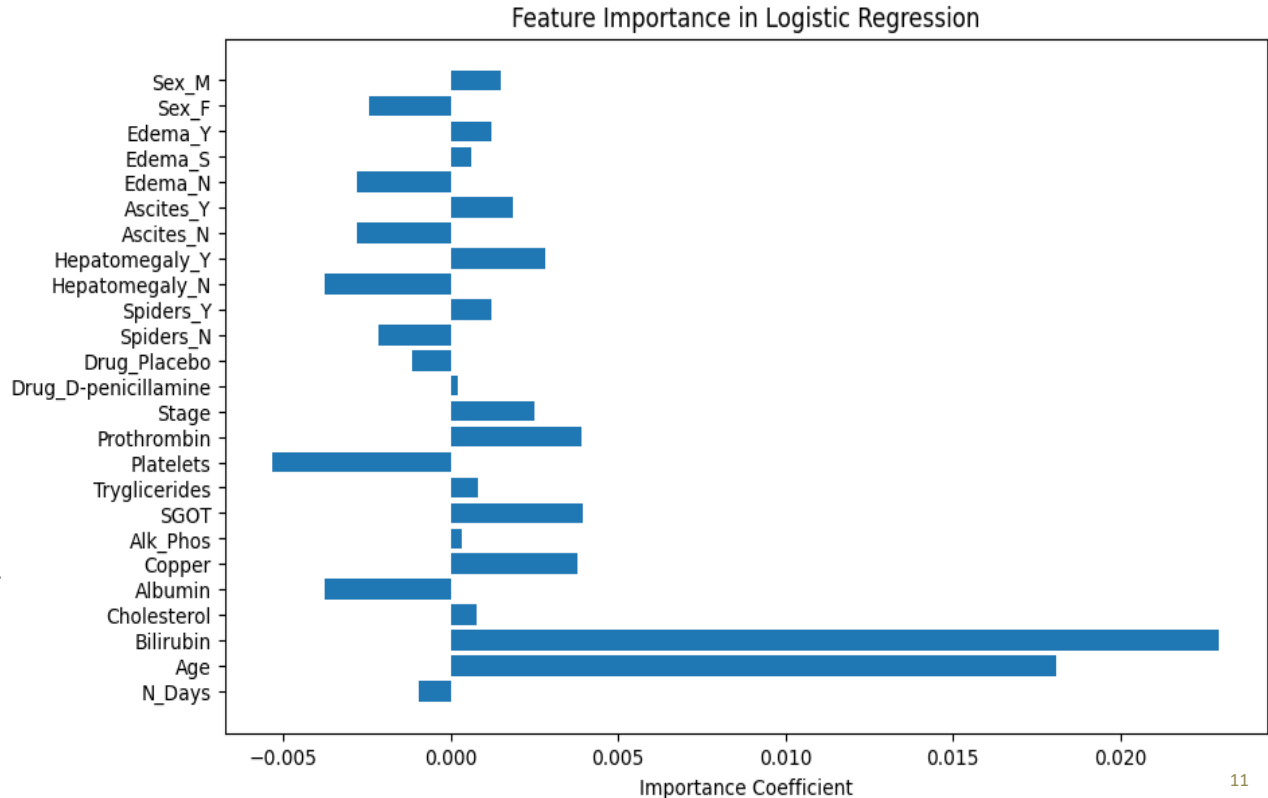
Table 4: Models Comparative Results

Model	All		Univariate		Multivariate		Wrapper	
	F-score	Brier-score	F-score	Brier-score	F-score	Brier-score	F-score	Brier-score
DT	0.6881 ± 0.06	0.1908 ± 0.03	0.6877 ± 0.06	0.1951 ± 0.04	0.6988 ± 0.05	0.1895 ± 0.02	0.6644 ± 0.05	0.2162 ± 0.03
KNN	0.7190 ± 0.07	0.1747 ± 0.02	0.7134 ± 0.07	0.1736 ± 0.02	0.7134 ± 0.07	0.1744 ± 0.02	0.7190 ± 0.07	0.1752 ± 0.02
RIPPER	0.5595 ± 0.11	0.2251 ± 0.05	0.5461 ± 0.09	0.2078 ± 0.03	0.6264 ± 0.07	0.1970 ± 0.03	0.6067 ± 0.07	0.1905 ± 0.02
kerasMLP	0.6485 ± 0.03	—	0.6586 ± 0.04	—	0.5983 ± 0.10	—	0.6900 ± 0.05	—
sklearnMLP	0.7081 ± 0.03	0.1543 ± 0.02	0.6945 ± 0.05	0.1611 ± 0.03	0.7012 ± 0.06	0.1660 ± 0.01	0.7109 ± 0.05	0.1547 ± 0.02
SVM	0.7715 ± 0.04	—	0.7604 ± 0.05	—	0.7340 ± 0.06	—	0.7244 ± 0.05	—
LR	0.7593 ± 0.05	0.1484 ± 0.02	0.7596 ± 0.06	0.1484 ± 0.02	0.7507 ± 0.04	0.1591 ± 0.02	0.7728 ± 0.04	0.1563 ± 0.02
LDA	0.7516 ± 0.04	0.1467 ± 0.01	0.7452 ± 0.06	0.1491 ± 0.02	0.7577 ± 0.04	0.1483 ± 0.01	0.7714 ± 0.05	0.1464 ± 0.01
QDA	0.7351 ± 0.07	0.1800 ± 0.04	0.7563 ± 0.08	0.1765 ± 0.05	0.7422 ± 0.02	0.1885 ± 0.03	0.6583 ± 0.09	0.2700 ± 0.13
GNB	0.6885 ± 0.03	0.2115 ± 0.03	0.7098 ± 0.03	0.1969 ± 0.03	0.7355 ± 0.04	0.1874 ± 0.02	0.7659 ± 0.05	0.1756 ± 0.04
RF	0.7497 ± 0.04	0.1360 ± 0.02	0.7461 ± 0.04	0.1388 ± 0.02	0.7589 ± 0.05	0.1355 ± 0.02	0.7881 ± 0.05	0.1389 ± 0.02
AB	0.7247 ± 0.06	0.1972 ± 0.01	0.7380 ± 0.08	0.1932 ± 0.01	0.7467 ± 0.04	0.1890 ± 0.01	0.7215 ± 0.09	0.2026 ± 0.01
Bagging	0.7521 ± 0.03	0.1462 ± 0.01	0.7340 ± 0.04	0.1437 ± 0.02	0.7332 ± 0.04	0.1408 ± 0.02	0.7586 ± 0.04	0.1495 ± 0.02
Stacking	0.7261 ± 0.06	0.1704 ± 0.03	0.7133 ± 0.06	0.1674 ± 0.02	0.7086 ± 0.07	0.1694 ± 0.02	0.7520 ± 0.04	0.1630 ± 0.03
Voting	0.7351 ± 0.04	—	0.7601 ± 0.05	—	0.7471 ± 0.04	—	0.7524 ± 0.06	—

* DT: Decision Tree, MLP: Multi Layer Perceptron; SVM: Support Vector Machine;
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Model Interpretability

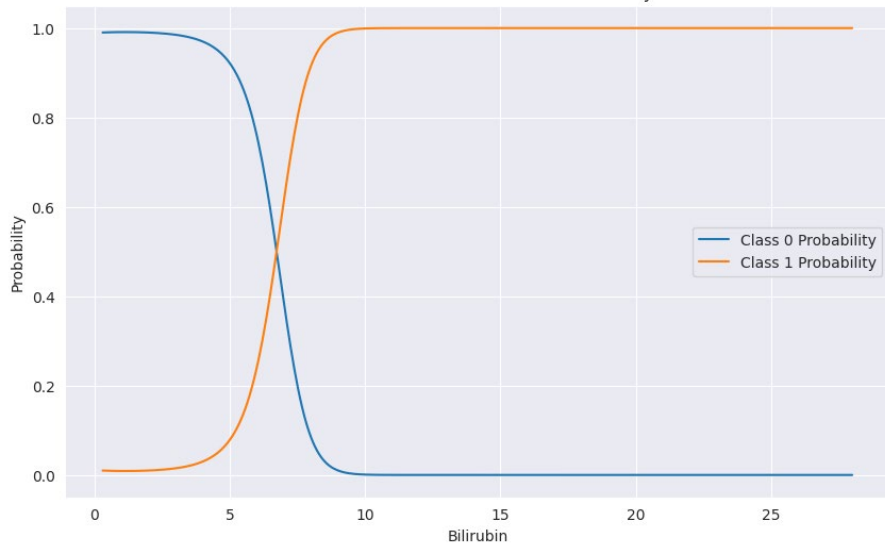
- Feature with a coefficient close to 0 indicates that the model does not consider the feature significant for decision-making (**Drug_D-penicillamine**).
- Positive coefficient, as **Age**, implying that as age increases, the likelihood of the model predicting death also increases.
- A high negative coefficient, as **platelets**, suggests a lower likelihood of being classified as deceased.



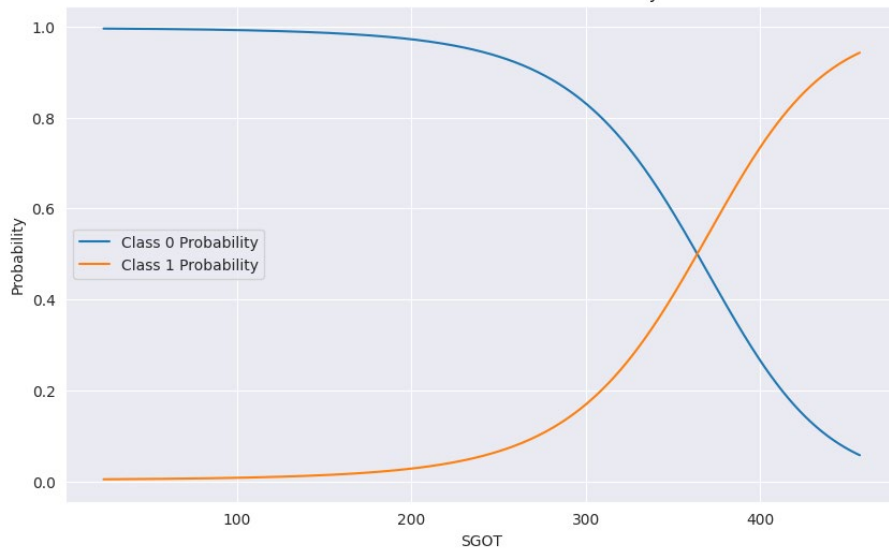
Model Interpretability

- Two examples wherein we take a single patient, alter one feature within the range from the minimum to maximum for that feature, while fixing the other features.
- From **SGOT** level of 200, the models begin to incrementally raise the probability of Death.
- **Bilirubin** feature, which exhibits a behavior similar but with a steeper curve.

Class Probabilities for Gaussian Naive Bayes



Class Probabilities for Gaussian Naive Bayes



Discussion & Conclusion

- The findings indicate that **SVM** achieves the highest F-score when employing all features and with the univariate subset.
- **Random Forest** performs best for the multivariate and wrapper subset, exhibiting the lowest Brier score, thereby demonstrating effective calibration.
- Model visualization present that **knowledge can be extracted** from Logistic Regression and Naïve Bayes.

Thanks!

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